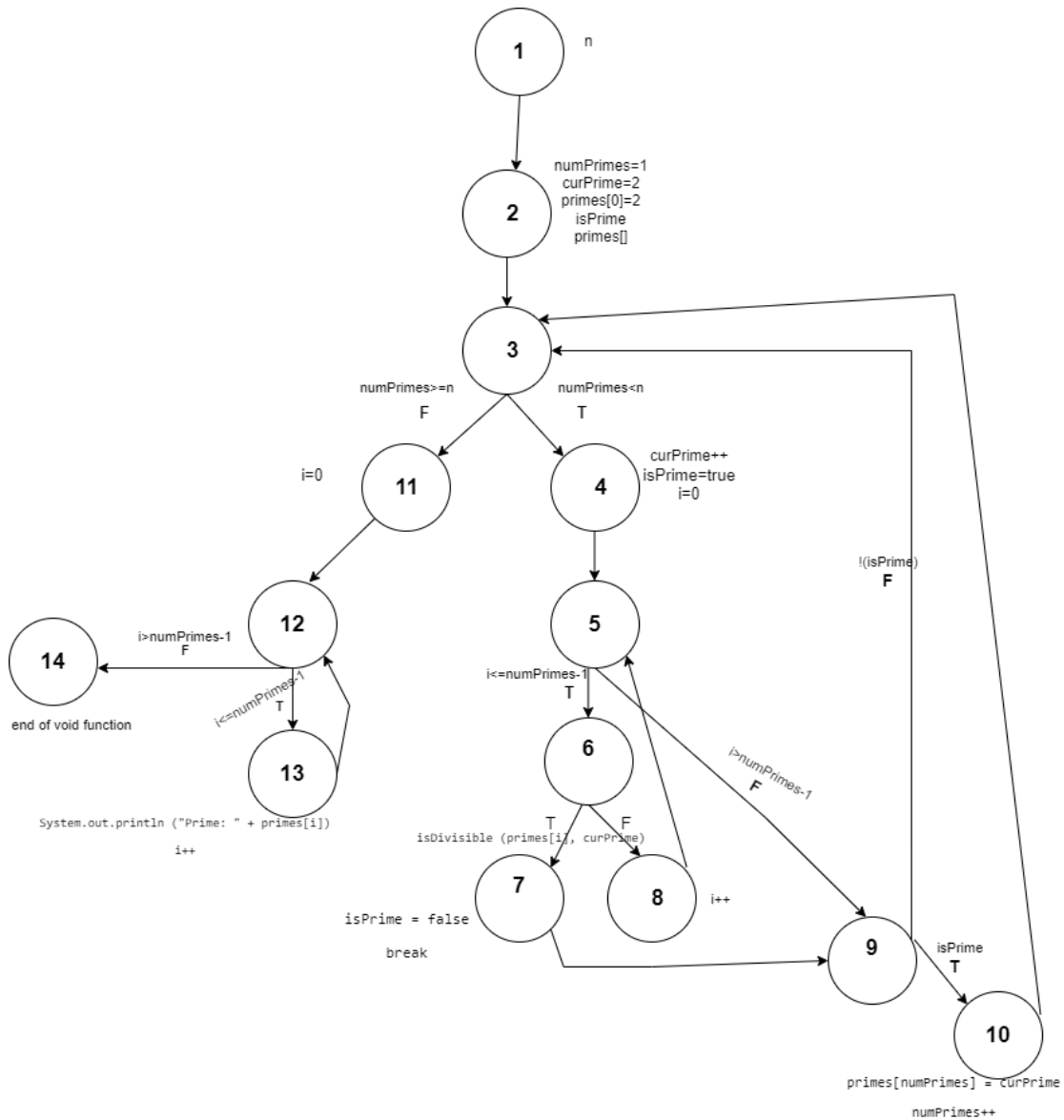


ДОМАШНА ЗАДАЧА 2

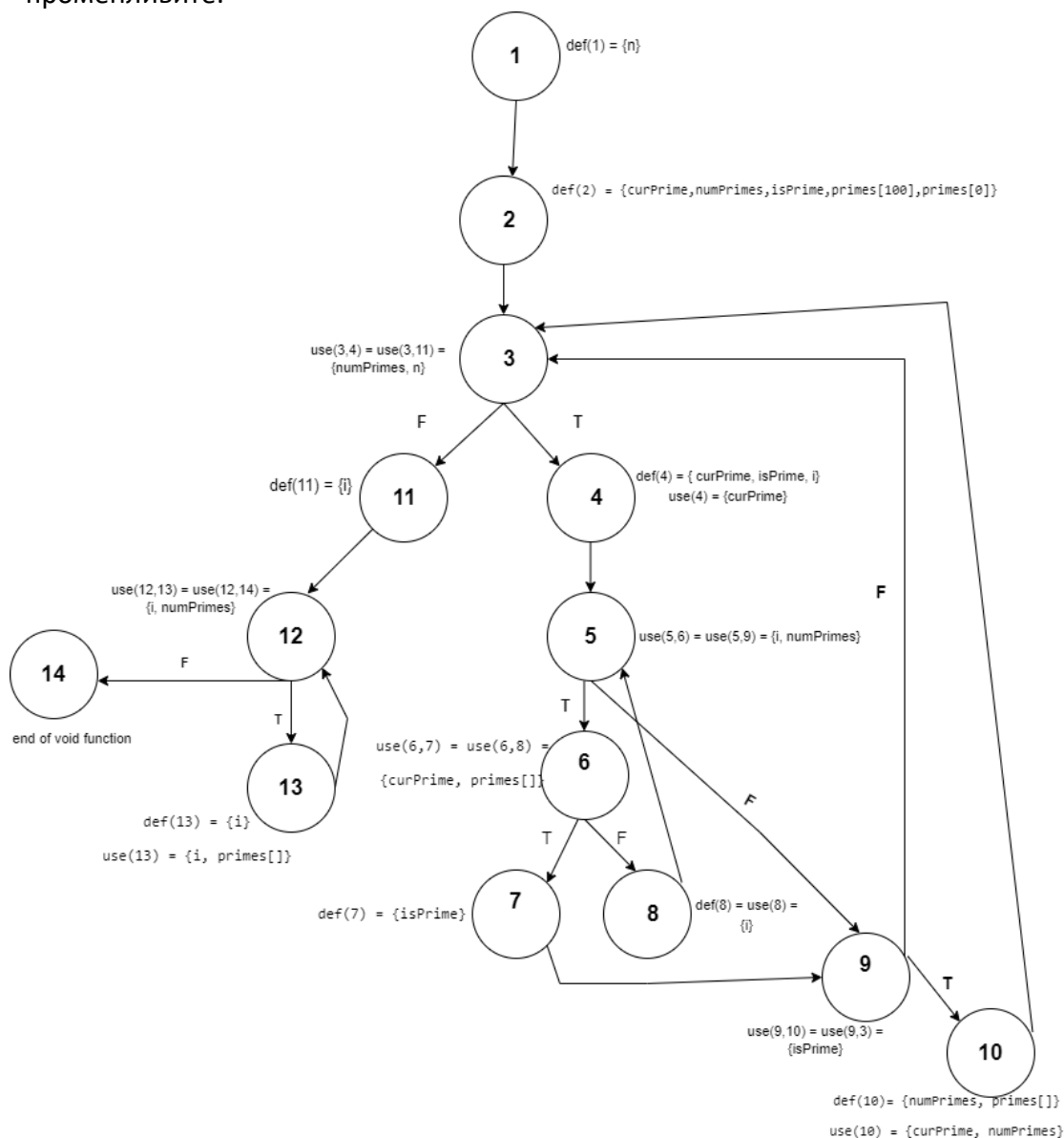
ПРЕДМЕТ: СОФТВЕРСКИ КВАЛИТЕТ И ТЕСТИРАЊЕ

СТУДЕНТ: Аритон Веруш

Податочниот тек на функцијата printPrimes прикажан со граф:



Графот со соодветните дефиниции и употреби (def-use, du-pairs) на променливите:



Соодветниот граф напишан во вид на табела за секоја променлива изгледа вака:

node	def	use
1	$\{n\}$	
2	$\{curPrime, isPrime, primes[], NumPrimes, primes[o]\}$	
3		$\{numPrimes, n\}$
4	$\{curPrime, isPrime, i\}$	$\{curPrime\}$
5		$\{i, numPrimes\}$
6		$\{curPrime, primes[]\}$
7	$\{isPrime\}$	
8	$\{i\}$	$\{i\}$
9		$\{isPrime\}$
10	$\{numPrimes, primes[]\}$	$\{curPrime, numPrimes\}$
11	$\{i\}$	
12		$\{i, numPrimes\}$

13	$\{i\}$	$\{i, \text{primes}[]\}$
14		

Следната табела ги прикажува сите def и use за секој edge посебно:

Edge	def	use
(1,2)	{n}	
(2,3)	{curPrime,numPrimes,isPrime, primes[],primes[0]}	
(3,4)		{numPrimes,n}
(3,11)		{numPrimes,n}
(4,5)	{curPrime,isPrime, i}	{curPrime}
(5,6)		{i,numPrimes}
(5,9)		{i,numPrimes}
(6,7)		{curPrime, primes[]}
(6,8)		{curPrime, primes[]}
(7,9)	{isPrime}	
(8,5)	{i}	{i}
(9,3)		{isPrime}
(9,10)		{isPrime}
(10,3)	{numPrimes,primes[]}	{curPrime, numPrimes}
(11,12)	{i}	
(12,13)		{i,numPrimes}
(12,14)		{i,numPrimes}
(13,12)	{i}	{i, primes[]}

Во следната табела се прикажани се сите du-paths за секој def и use на променлива посебно:

Variable	Du-path set	Du-paths	prefix
<i>n</i>	du(1, n)	[1, 2, 3]	
<i>curPrime</i>	du(2, curPrime) du(4, curPrime) du(4, curPrime) du(4, curPrime) du(4, curPrime) du(4, curPrime) du(4, curPrime) du(4, curPrime)	[2,3,4] [4,5,6] [4,5,9,10] [4,5,9,3,4] [4,5,9,10,3,4] [4,5,6,7,9,10] [4,5,6,7,9,3,4] [4,5,6,7,9,10,3,4]	Yes Yes Yes
<i>numPrimes</i>	du(2, numPrimes) du(2, numPrimes) du(2, numPrimes) du(2, numPrimes) du(2, numPrimes) du(10, numPrimes) du(10, numPrimes) du(10, numPrimes) du(10, numPrimes)	[2,3] [2,3,4,5] [2,3,11,12] [2,3,4,5,9,10] [2,3,4,5,6,7,9,10] [10,3] [10,3,4,5] [10,3,11,12] [10,3,4,5,9,10]	Yes Yes Yes Yes

	du(10, numPrimes)	[10,3,4,5,6,7,9,10]	
<i>isPrime</i>	du(2, isPrime) du(2, isPrime) du(4, isPrime) du(7, isPrime)	[2,3,4,5,9] [2,3,4,5,6,7,9] [4,5,9] [7,9]	Yes Yes
<i>primes[]</i>	du(2, primes[]) du(2, primes[]) du(10, primes[]) du(10, primes[])	[2,3,11,12,13] [2,3,4,5,6] [10,3,4,5,6] [10,3,11,12,13]	
<i>i</i>	du(11, i) du(11, i) du(11, i) du(13, i) du(13, i) du(4, i) du(4, i) du(8, i) du(4, i) du(4, i) du(4, i) du(4, i) du(4, i) du(4, i) du(4, i) du(8, i) du(8, i) du(8, i) du(8, i)	[11, 12] [11, ,12 ,13] [11, 12, 13, 12] [13. 12] [13, 12, 13] [4, 5] [4, 5, 6, 8] [8, 5] [4, 5, 9, 3, 11 ,12] [4, 5, 9, 3, 11, 12, 13, 12, 14] [4, 5, 9, 10, 3, 11 ,12] [4, 5, 9, 10, 3, 11, 12, 13, 12,14] [4, 5, 6, 7, 9, 3, 11, 12, 13, 12,14] [4, 5, 6, 7, 9, 10 , 3, 11, 12, 13, 12, 14] [8, 5, 6, 7, 9, 3, 11, 12, 13, 12, 14] [8, 5, 6, 7, 9, 10, 3, 11, 12, 13, 12, 14] [8, 5, 9, 3, 11, 12, 13, 12, 14] [8, 5, 9, 10, 3, 11, 12, 13, 12, 14]	Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes Yes
<i>primes[0]</i>	du(9, primes[0])	No path or No path needed	

Соодветните тестови со тест патиштата:

Test case (n)	Test paths that satisfy all du-path coverage
(n<=1, void return) If we are searching for 1 prime number or less, given that number 2 is already a prime number we have the following path:	[1,2,3,11,12,13,12,14]
(n=2 , void return) If we are searching for 2 prime numbers, with number 2 already being counted as a prime number, we end up finding that number 3, who is next to number 2 is also a prime number:	[1,2,3,4,5,6,8,5,9,10,3,11,12,13,12,14]
(n>2, void return) If we are searching for more than 2 prime numbers, we end up in cycles with 2 different paths(the number is or the number isn't a prime number) until we reach the number of prime numbers we want having the following test path:	[1,2,3,4,5,6,8,5,9,10,3,4,5,6,7,9,3,4,5,6,8,5,9,10,3,11,12,13,12,14]